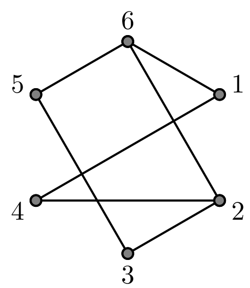
Math 252 B
Spring 2022

Name ________________________________ Section: A B

Final Exam

1 Answer the following questions about the graph below.



a) What is an Euler circuit?

b) Does the graph have an Euler circuit? How do you know?

c) Does the graph have an Euler path that is not a circuit? How do you know?

d) Is the graph bipartite? How do you know?

e) Is the graph planar? How do you know?

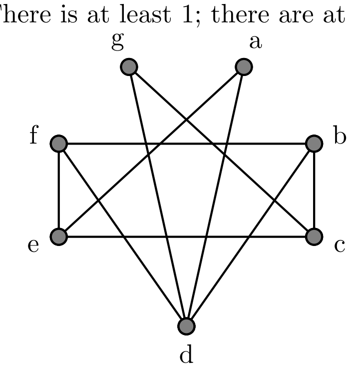
2 Draw a simple graph with 7 vertices, each of which has degree 4.

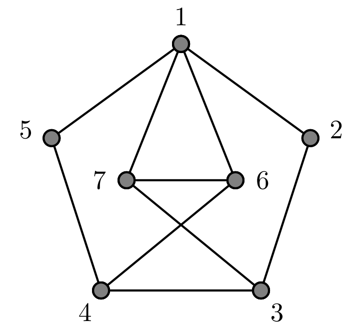
3 Explain why it is impossible to draw a simple graph with 7 vertices, each of degree 3.

4 Give an example of a graph that is **not a planar graph** and explain how you know it is not planar. **Do not use Kuratowski's Theorem.**

5 How many isomorphisms are there between the following two graphs? **Explain.**

(If you do not know exactly how many there are, tell me as much as you know about how many there are. Examples: There is at least 1; there are at most 3, etc.)





6

a) Show how to compute $\binom{11}{5}$ **using only basic calculator operations** (addition, subtraction, multiplication, and division).

b) Write a counting problem for which $\binom{11}{5}$ is the answer. [Keep it simple.]

7 $\mathrm{P}(A) = 0.4$, $\mathrm{P}(B) = 0.5$, and $\mathrm{P}(A \cap B) = 0.15$. Use this to answer the following questions. Be sure to show your work.

a) What is $\mathrm{P}(A \cup B)$?

b) What is $\mathrm{P}(A \mid B)$?

c) Are A and B mutually exclusive? (Explain your answer.)

d) Are A and B independent? (Explain your answer.)

8 4's on Four 4-Sided Dice. Let X be the number of 4's when rolling four 4-sided dice.

a) What is $\mathrm{P}(X = 0)$?

b) What is $\mathrm{P}(X = 2)$?

c) What is the expected value of X? (This is denoted $\mathrm{E}(X)$.)

9 A company orders parts from three different suppliers.

- 70% come from supplier A,
- 25% come from supplier B, and
- 5% come from supplier C.

Some of the parts are defective.

- 2% of the parts from supplier A are defective,
- 4% of the parts from supplier B are defective,
- 5% of the parts from supplier C are defective.

a) Altogether, what percent of the ordered parts are defective?

b) What percent of the defective parts come from supplier A?

10 Dice.

a) If you roll three dice (each with six sides numbered 1, 2, 3, 4, 5, and 6), what is the probability that the three numbers rolled will all be the same?

b) If you roll three dice (each with six sides numbered 1, 2, 3, 4, 5, and 6), what is the probability that the three numbers rolled will be distinct? (no repetitions)

11 What is the output of the following python code?

```python
import re as re
re.findall(r'a+b*[cd]*', 'abcdabcddcbaabbccddccbbaa')
```

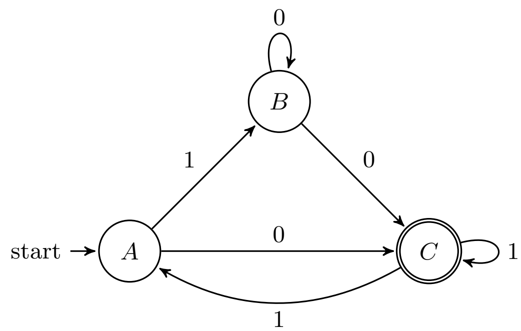
12 Let L be the langauge consisting of all binary strings that end with at least two 0's followed by at least one 1. Examples: $10011 \in L$, $0010 \notin L$, $01011 \notin L$.

a) Write a **regular expression** for L.

b) Construct a **DFA** that recognizes L.

13 Give an example of a language that is **not regular**, and explain how you know the language is not regular.

14 Below is the state diagram for an **NFA** N.



a) Which of the following strings are accepted by N? (Simply write "accept" or "reject" after each.)

- 101
- 0100
- 0101
- 0011
- 1001
- 01100

b) Construct a **DFA** that accepts the same language as N using a **general method** that works for all NFAs.

c) Construct a **regular grammar** for this language using a **general method** that works for all NFAs.

15 Here is a Turing machine program using the syntax of the simulator we used in class. The start state is 0 and the input alphabet is $\{a, b\}$.

```
0 a b R 1
0 b a R 2

1 a a L 0
1 b b L 3

2 a b R 3
2 b a R 0

3 a a L 2
3 b b L 1
```

Suppose we run this machine with input *aabaa*. Simulate the Turing machine **for 5 steps or until it halts**, whichever comes first.

a) Write down the state the machine is in *after each step.*

b) Write down the the tape contents *at the end of your simulation.*

c) Use an arrow to indicate the location of the read/write head at the end of your simulation.

States: 0 ______ ______ ______ ______ ______

Tape Contents (at end of simulation, arrow pointing to read/write head position):

…													…